

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	58980	52.260363
1	[Git, Change, Create]	17328	15.353807
2	[Git, Commit]	11315	10.025873
3	[Git, Change, Delete]	8787	7.785890
4	[File, Git]	5920	5.245530
5	[Git, Change, Rename]	3354	2.971876
6	[Git, Tag]	1795	1.590494
7	[Author, Git, Person]	1268	1.123536
8	[Json, Key]	668	0.591894
9	[Json, Value, Scalar]	603	0.534300
10	[Committer, Git, Person]	370	0.327846
11	[NPM, Dependency]	338	0.299491
12	[Type, TS, Primitive]	293	0.259618
13	[Type, TS, Declared]	277	0.245441
14	[TS, ExternalDeclaration]	215	0.190505
15	[Git, Change, Copy]	138	0.122278
16	[Type, TS, Literal]	136	0.120505
17	[Json, Value, Object]	133	0.117847
18	[Type, TS, Union]	120	0.106328
19	[Type, TS, ObjectMember]	103	0.091265
20	[NPM, Script]	91	0.080632
21	[TS, Property]	65	0.057594
22	[TS, Function]	47	0.041645
23	[Git, Branch]	45	0.039873
24	[Type, TS, FunctionParameter]	40	0.035443
25	[Type, Object, TS]	39	0.034557
26	[File, Directory]	34	0.030126
27	[Type, TS, Function]	34	0.030126
28	[TS, Parameter]	33	0.029240
29	[Package, File, Json, NPM]	29	0.025696

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

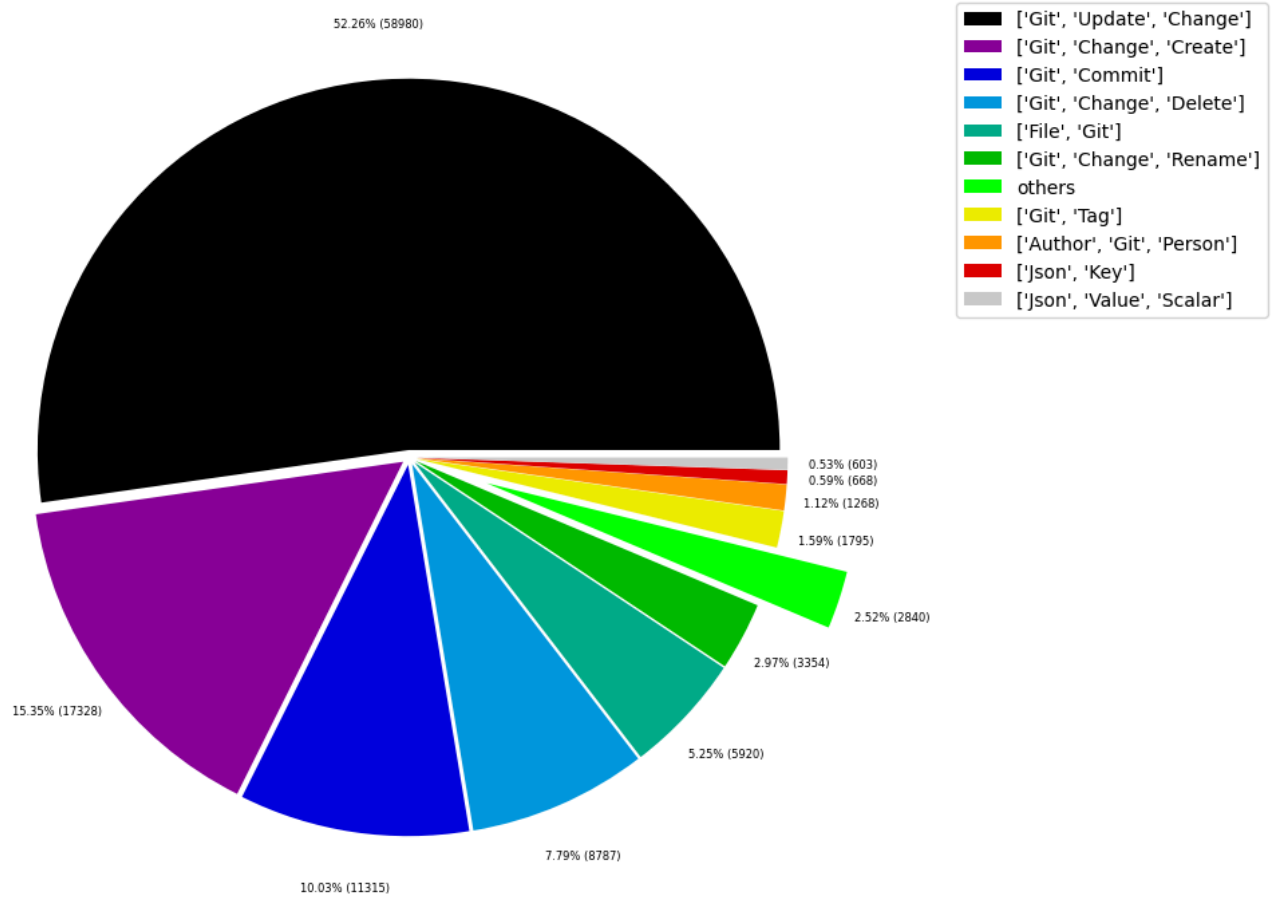


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.000886
1	[NPM, PublishConfig]	1	0.000886
2	[File, TS, Scan]	1	0.000886
3	[TS, Method]	1	0.000886
4	[Repository, File, Git]	1	0.000886
5	[TS, Constructor]	1	0.000886
6	[Value, TS, ObjectMember]	1	0.000886
7	[TS, Class]	1	0.000886
8	[TS, Enum]	2	0.001772
9	[Value, Object, TS]	3	0.002658
10	[Type, TS, Tuple]	3	0.002658
11	[Value, TS, Function]	4	0.003544
12	[TS, TypeParameter]	4	0.003544
13	[Value, TS, Complex]	5	0.004430
14	[Repository, NPM]	5	0.004430
15	[NPM, Engine]	6	0.005316
16	[Project, TS]	6	0.005316
17	[File, Local]	6	0.005316
18	[Type, TS, TypeParameterReference]	6	0.005316
19	[Value, TS, Member]	6	0.005316
20	[File, TS, Local, Module]	6	0.005316
21	[Value, TS, Call]	6	0.005316
22	[TS, EnumMember]	8	0.007089
23	[Type, TS, NotIdentified]	11	0.009747
24	[TS, ExternalModule]	11	0.009747
25	[Json, Value, Array]	12	0.010633
26	[Value, TS, Declared]	13	0.011519
27	[TS, TypeAlias]	16	0.014177
28	[File, Directory, Local]	16	0.014177
29	[TS, Interface]	17	0.015063

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (less than 0.5% overall)

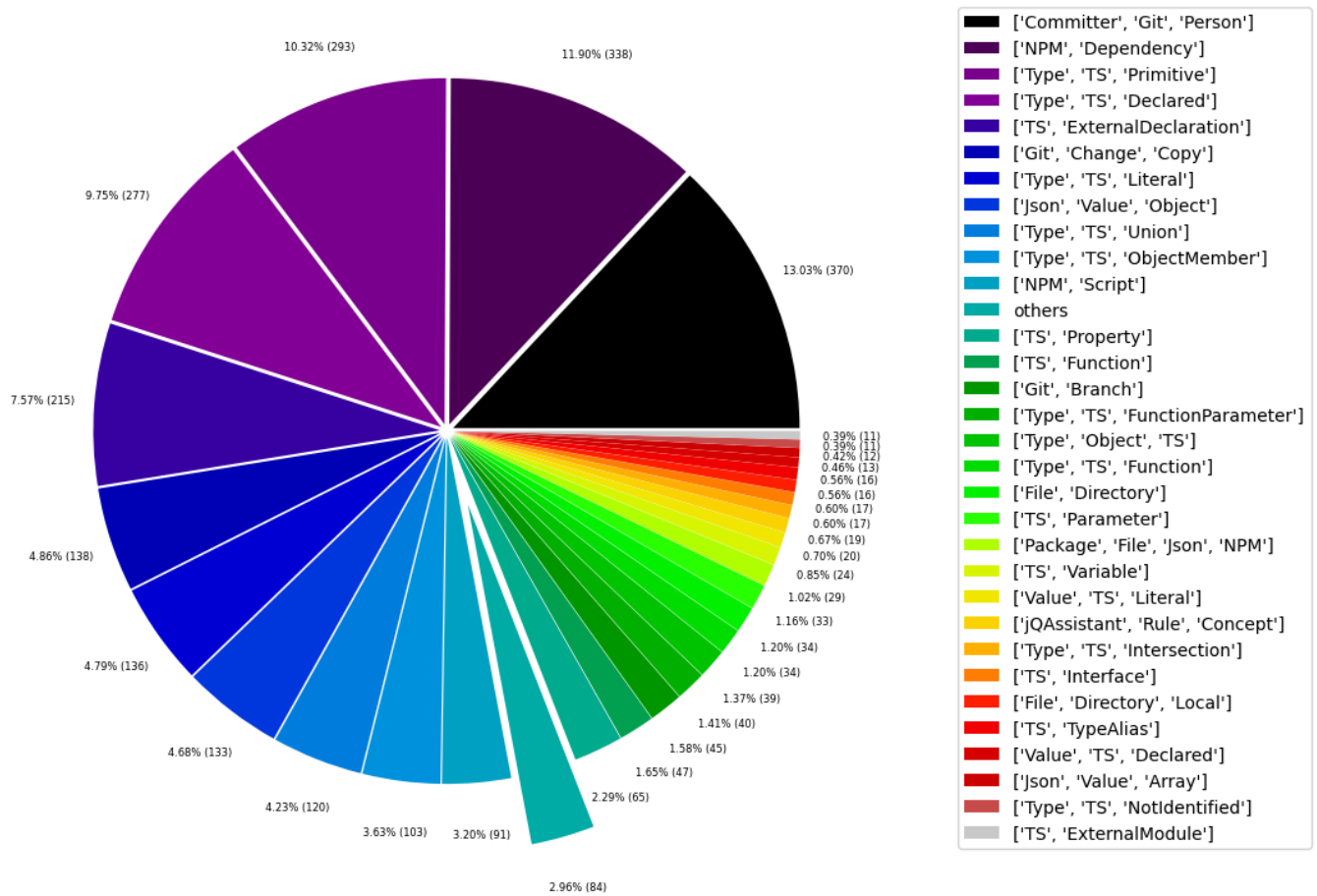


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

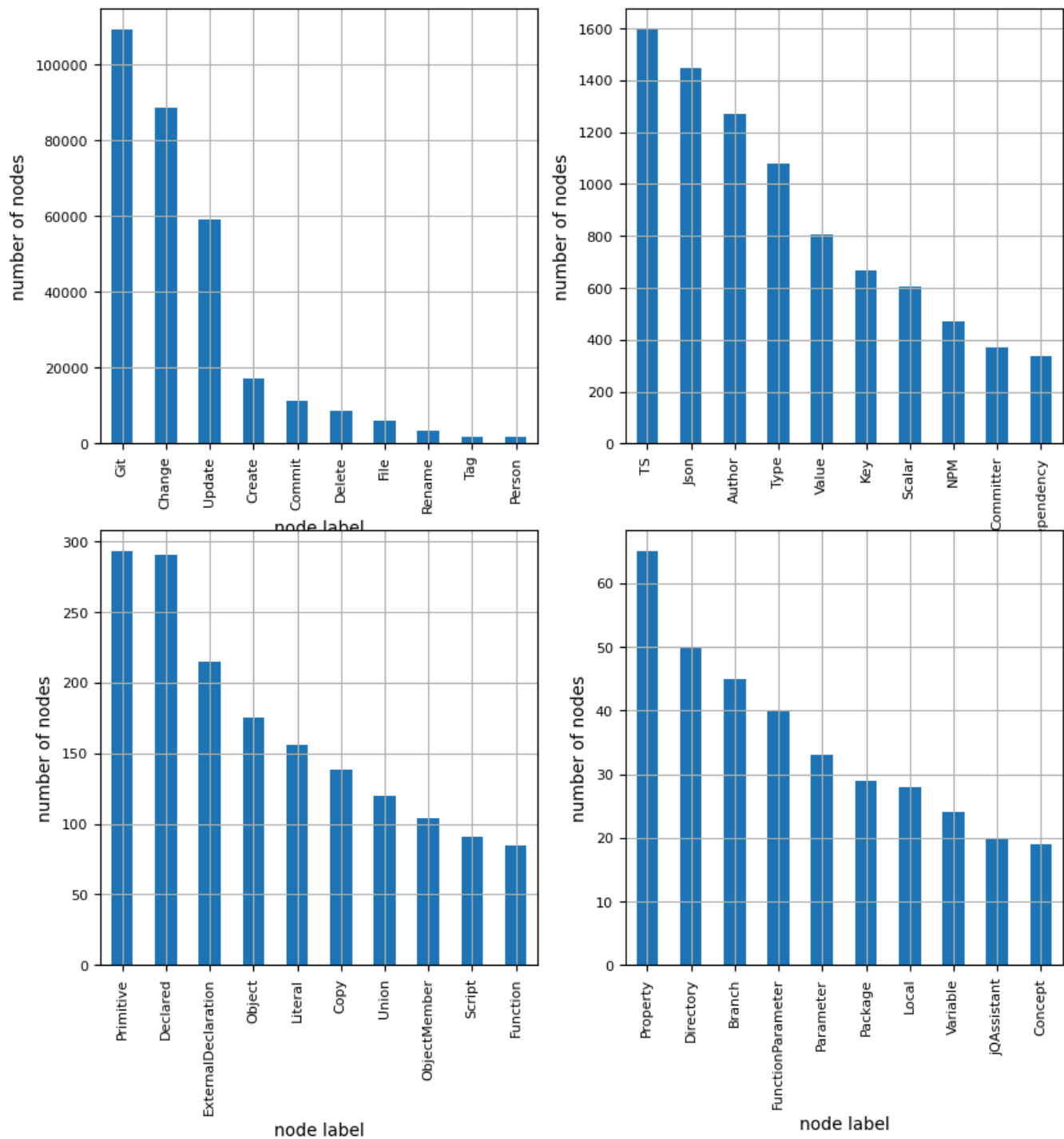
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	109301	96.848252
1	Change	88587	78.494214
2	Update	58980	52.260363
3	Create	17328	15.353807
4	Commit	11315	10.025873
5	Delete	8787	7.785890
6	File	6013	5.327934
7	Rename	3354	2.971876
8	Tag	1795	1.590494
9	Person	1638	1.451381
10	TS	1595	1.413280
11	Json	1445	1.280370
12	Author	1268	1.123536
13	Type	1079	0.956069
14	Value	806	0.714172
15	Key	668	0.591894
16	Scalar	603	0.534300
17	NPM	470	0.416453
18	Committer	370	0.327846
19	Dependency	338	0.299491
20	Primitive	293	0.259618
21	Declared	290	0.256960
22	ExternalDeclaration	215	0.190505
23	Object	175	0.155062
24	Literal	156	0.138227
25	Copy	138	0.122278
26	Union	120	0.106328
27	ObjectMember	104	0.092151
28	Script	91	0.080632
29	Function	85	0.075316
30	Property	65	0.057594
31	Directory	50	0.044303
32	Branch	45	0.039873
33	FunctionParameter	40	0.035443
34	Parameter	33	0.029240
35	Package	29	0.025696
36	Local	28	0.024810
37	Variable	24	0.021266
38	jqAssistant	20	0.017721
39	Concept	19	0.016835

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

<Figure size 640x480 with 0 Axes>

Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 336953

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	88587	26.290610
1	MODIFIES	88587	26.290610
2	UPDATES	58980	17.503925
3	COMMITTED	22630	6.716070
4	CREATES	20820	6.178903
5	HAS_PARENT	12401	3.680335
6	DELETES	12141	3.603173
7	HAS_COMMIT	11315	3.358035
8	HAS_FILE	5920	1.756922
9	RENAMES	3354	0.995391
10	HAS_TAG	1795	0.532715
11	ON_COMMIT	1795	0.532715
12	HAS_NEW_NAME	1770	0.525296
13	HAS_AUTHOR	1268	0.376314
14	DEPENDS_ON	887	0.263241
15	HAS_KEY	668	0.198247
16	HAS_VALUE	668	0.198247
17	CONTAINS	596	0.176879
18	HAS_COMMITTER	370	0.109808
19	OF_TYPE	339	0.100608
20	EXPORTS	309	0.091704
21	REFERENCES	197	0.058465
22	DECLARES	186	0.055201
23	DECLARES_DEV_DEPENDENCY	169	0.050155
24	DECLARES_DEPENDENCY	161	0.047781
25	COPIES	138	0.040955
26	HAS_MEMBER	104	0.030865
27	HAS_TYPE_ARGUMENT	94	0.027897
28	DECLARES_SCRIPT	91	0.027007
29	RETURNS	82	0.024336

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

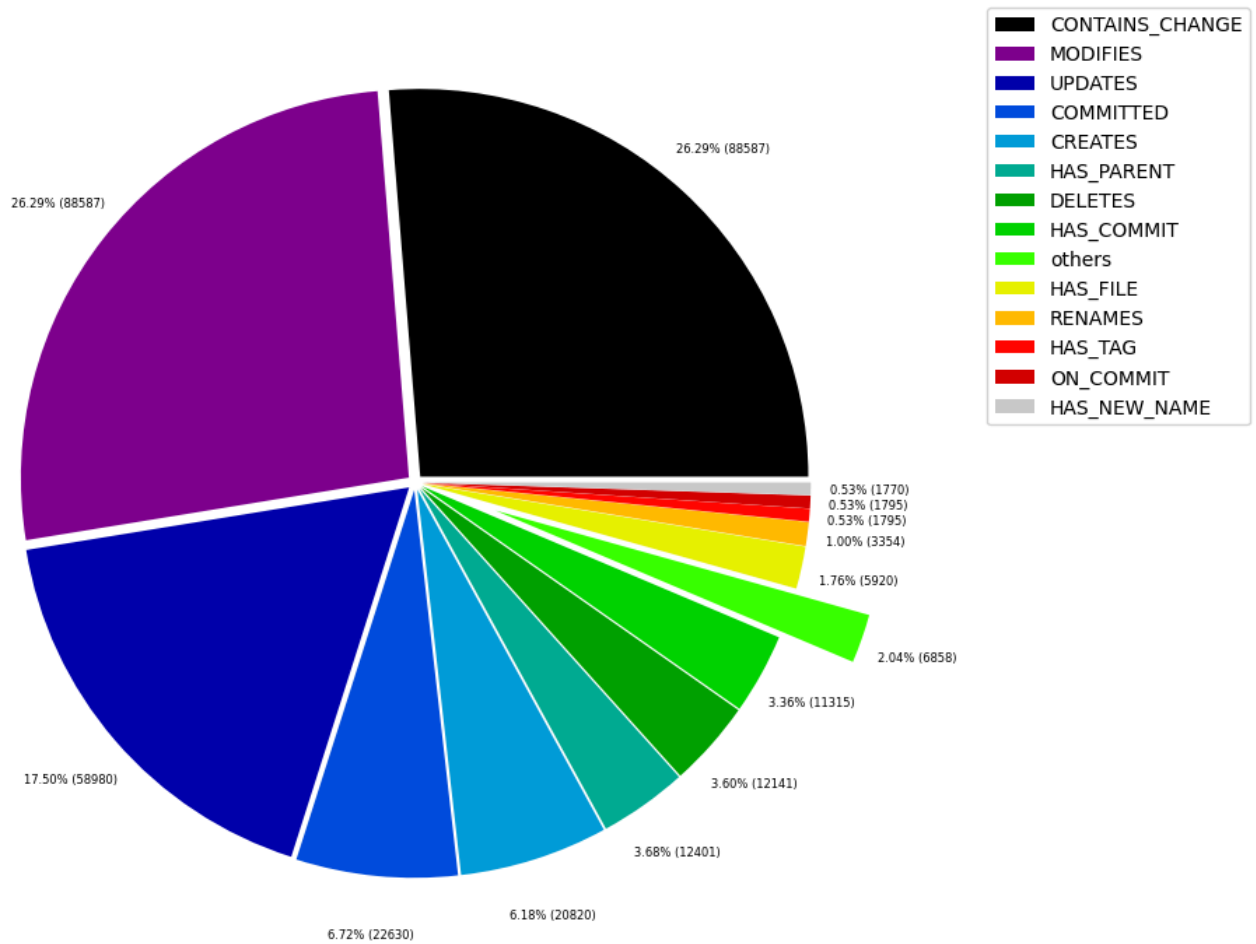


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

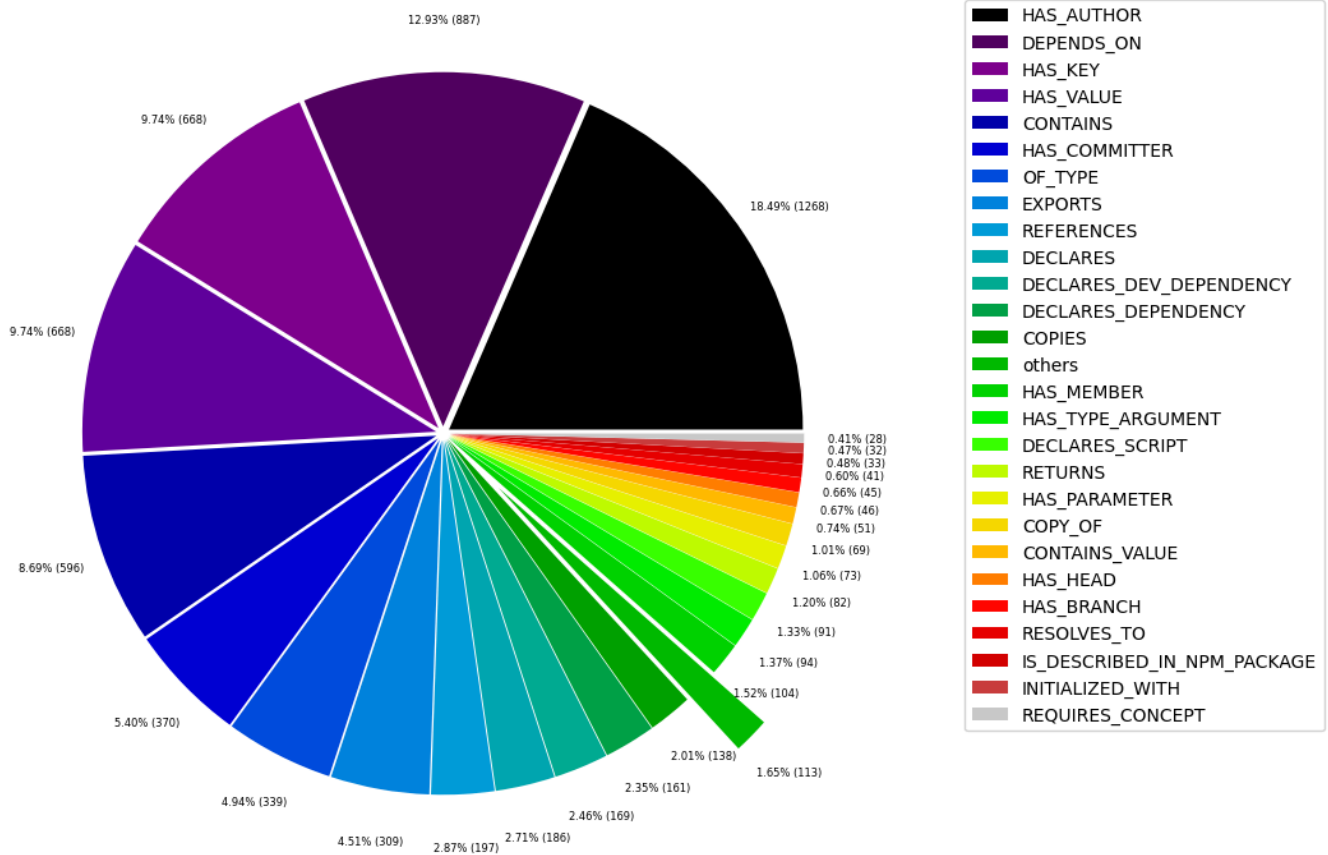
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	DECLARES_PUBLISH_CONFIG	1	0.000297
1	CONSTRAINED_BY	4	0.001187
2	IN_REPOSITORY	5	0.001484
3	REFERENCED_PROJECTS	5	0.001484
4	PARENT	6	0.001781
5	MEMBER	6	0.001781
6	HAS_ROOT	6	0.001781
7	HAS_NPM_PACKAGE	6	0.001781
8	HAS_CONFIG	6	0.001781
9	HAS_ARGUMENT	6	0.001781
10	EXTENDS	6	0.001781
11	DECLARES_ENGINE	6	0.001781
12	CONTAINS_PROJECT	6	0.001781
13	CALLS	6	0.001781
14	DECLARES_PEER_DEPENDENCY	8	0.002374
15	USES	11	0.003265
16	INCLUDES_CONCEPT	19	0.005639
17	REQUIRES_CONCEPT	28	0.008310
18	INITIALIZED_WITH	32	0.009497
19	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.009794
20	RESOLVES_TO	41	0.012168
21	HAS_BRANCH	45	0.013355
22	HAS_HEAD	46	0.013652
23	CONTAINS_VALUE	51	0.015136
24	COPY_OF	69	0.020478
25	HAS_PARAMETER	73	0.021665
26	RETURNS	82	0.024336
27	DECLARES_SCRIPT	91	0.027007
28	HAS_TYPE_ARGUMENT	94	0.027897
29	HAS_MEMBER	104	0.030865

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOf
0	[Git, Update, Change]	MODIFIES	[File, Git]	58980	58980	
1	[Git, Update, Change]	UPDATES	[File, Git]	58980	58980	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	58980	11315	
3	[Git, Change, Create]	MODIFIES	[File, Git]	17328	17328	
4	[Git, Change, Create]	CREATES	[File, Git]	17328	17328	
5	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	17328	11315	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	12401	11315	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11315	1	
8	[Author, Git, Person]	COMMITTED	[Git, Commit]	11315	1268	
9	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11315	370	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	8787	11315	
11	[Git, Change, Delete]	MODIFIES	[File, Git]	8787	8787	
12	[Git, Change, Delete]	DELETES	[File, Git]	8787	8787	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	5920	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	3354	11315	
15	[Git, Change, Rename]	RENAMES	[File, Git]	3354	3354	
16	[Git, Change, Rename]	MODIFIES	[File, Git]	3354	3354	
17	[Git, Change, Rename]	DELETES	[File, Git]	3354	3354	
18	[Git, Change, Rename]	CREATES	[File, Git]	3354	3354	
19	[Repository, File, Git]	HAS_TAG	[Git, Tag]	1795	1	
20	[Git, Tag]	ON_COMMIT	[Git, Commit]	1795	1795	
21	[File, Git]	HAS_NEW_NAME	[File, Git]	1770	5920	
22	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	1268	1	
23	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
24	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
25	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	370	1	
26	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
27	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
28	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	
29	[Package, File, Json, NPM]	DECLARES_DEV_DEPENDENCY	[NPM, Dependency]	169	29	

Graph Density

```

total_number_of_nodes (vertices): 112858
total_number_of_relationships (edges): 336953
-> total directed graph density: 2.645504326668455e-05
-> total directed graph density in percent: 0.0026455043266684552

```